MASDA: A Multi-scale Affective Systems & Disorder Atlas

– Hendrik Willems –

The Problem: "Scale Gap"

Affective neurobiology is fragmented; clinical diagnoses lack direct, programmable links to cellular taxonomies.

The Scope

11 disorders – 131 brain regions – 461 cells – 178 genes – 53 pharmaceuticals

Goals:

To engineer a relational discovery engine that projects macro-scale psychiatric dysfunction onto micro-scale ion channel profiles.

Sources

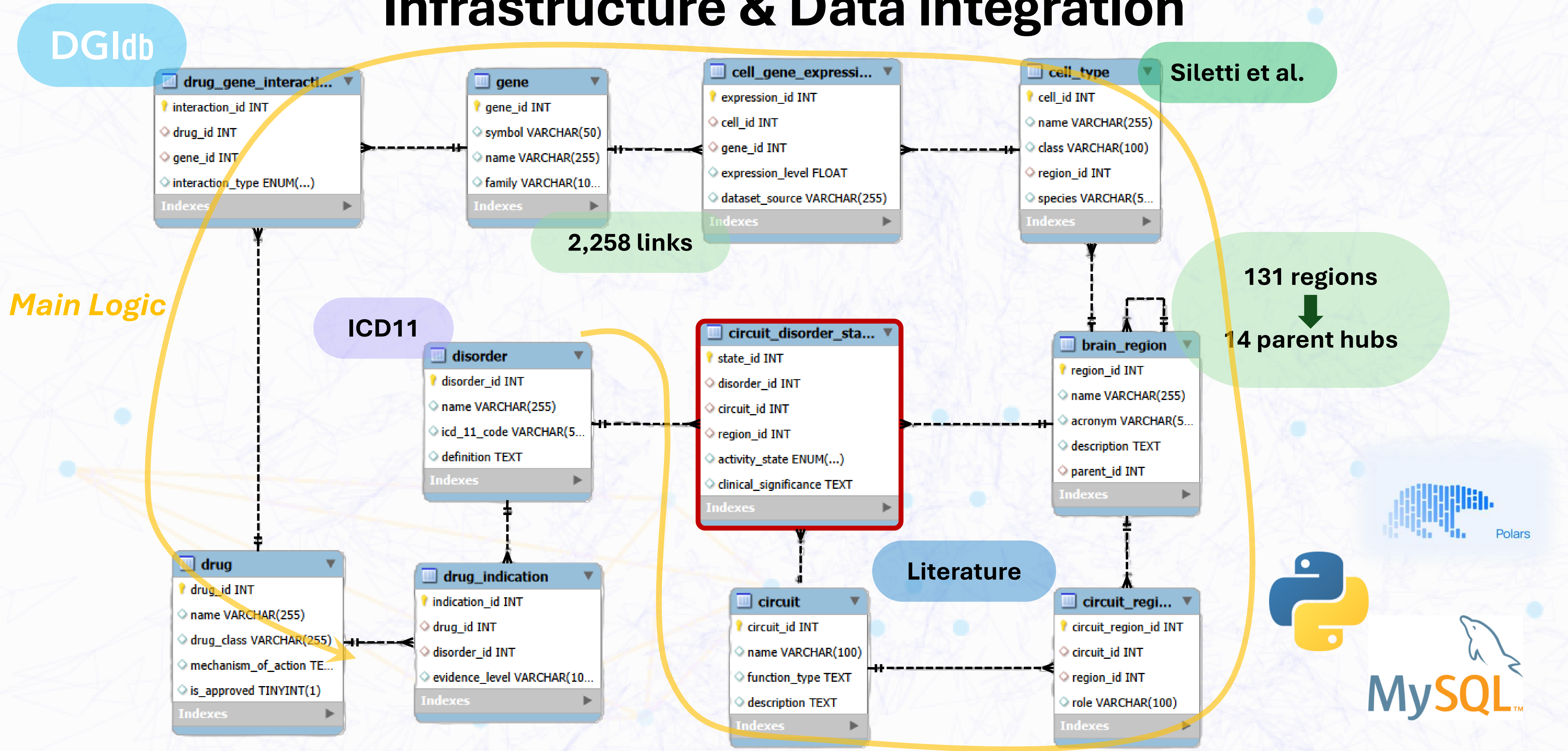
LITERATURE

WHO/ICD-11

DGIdb

ALLEN INSTITUTE *for* BRAIN SCIENCE

Infrastructure & Data integration



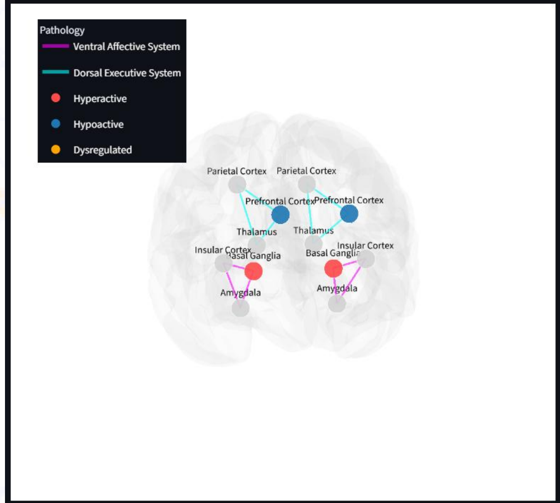
state_id	disorder_id	circuit_id	region_id	activity_state	clinical_significance
1	4	9	119	Hyperactive	Overactive to ...

Analysis and Interface

MASDA: From Disorder to Channel

Clinical Profile: Bipolar type I disorder, unspecified

ICD-11 Code: 6A60.Z
Clinical Definition: A mood disorder marked by extreme oscillations between mania and depression. Mania is scientifically linked to hyper-responsivity in the ventral striatum (reward system) and a lack of stabilization from the prefrontal cortex.



1. Anatomical Hub

Focus Parent Region

Basal Ganglia

Involved in reward processing, motor control, and the habit-forming CSTC loops.

Pathology & Evidence: Ventral striatal compartment hyperactivation drives reward-hypersensitivity and impulsivity during manic phases. (Phillips et al., 2003)

2. Cellular Population

Resident Cluster

Splat_403 (Inhibitory) - nucleus accumbens

3. Putative Molecular Targets & Therapeutics

Target Ion Channel

SCN1A

Therapeutics for Bipolar type I disorder, unspecified targeting SCN1A:

	name	is_approved
0	TOPIRAMATE	True
1	ZONISAMIDE	True
2	CARBAMAZEPINE	True
3	LEVETIRACETAM	True
4	LAMOTRIGINE	True